

Residence Radar

Time Series Forecasting Application

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Use Case

When people are looking for a new house ...



**Which
neighborhood
to live in**



**Best
forecasted
house price**



**Potential crime
rate**



**Income
Demographics**

App Functions

For users/home seekers

- Which neighborhood to live
- How expensive the houses would be
- how safe, the income levels in any neighborhood
- Suitable houses for sale and realtors

For advertisers/realtors and sellers

- Advertise the house for sale
- Get new clients

For builders/city planners

- Where to build houses - need our usage data

Data

Median income

Median income from 2011
to 2022 in Pittsburgh
by zip code.

Crime rate

Crime rate from 2018 to
2023 in Pittsburgh
by zip code.

House price

Median
House price from 2000 to
2023 in Pittsburgh
by zip code.



Values & Benefits

Value Proposition

- Users make more informed decisions
- Concrete data

Product Differentiation

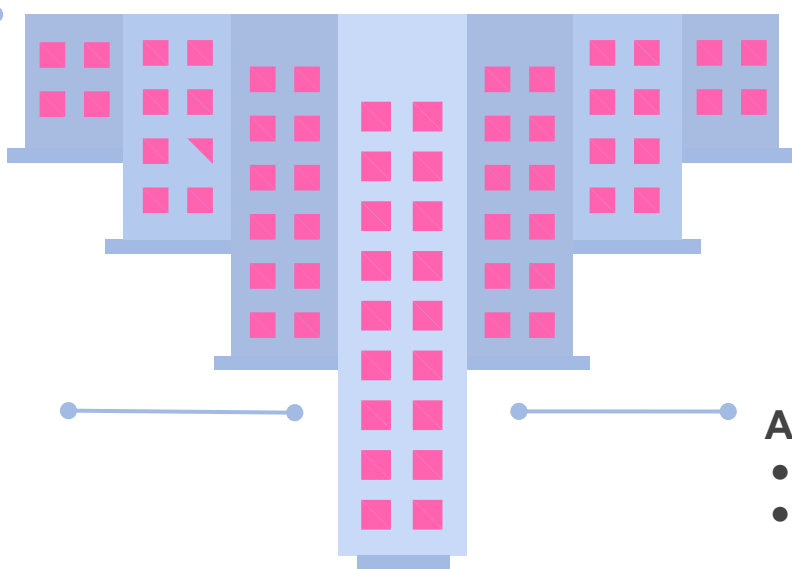
- Application streamlines
- Essential data
- User-friendly interface
- Trustworthy & valuable resource

Economic Benefits

- Time savings: all information in one place.
- Money savings: informed decisions, avoid costly mistakes.

Additional data for Full Value

- Time series data
- Safe cloud storage data



Discovery Hypothesis

CUSTOMER PROBLEM: We believe that individuals seeking housing in Pittsburgh face the challenge of accessing information about specific zip code areas.

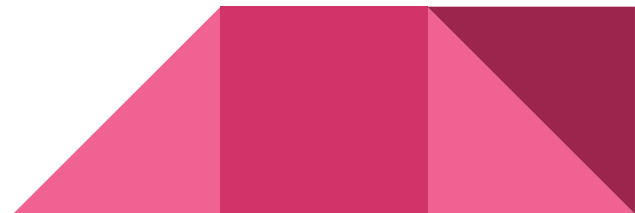
PROPOSED SOLUTION: We propose to address this problem by developing an intuitive application that allows users to input a specific zip code and instantly access detailed data relevant to that area.

HYPOTHESES UNDERLYING THE IDEA

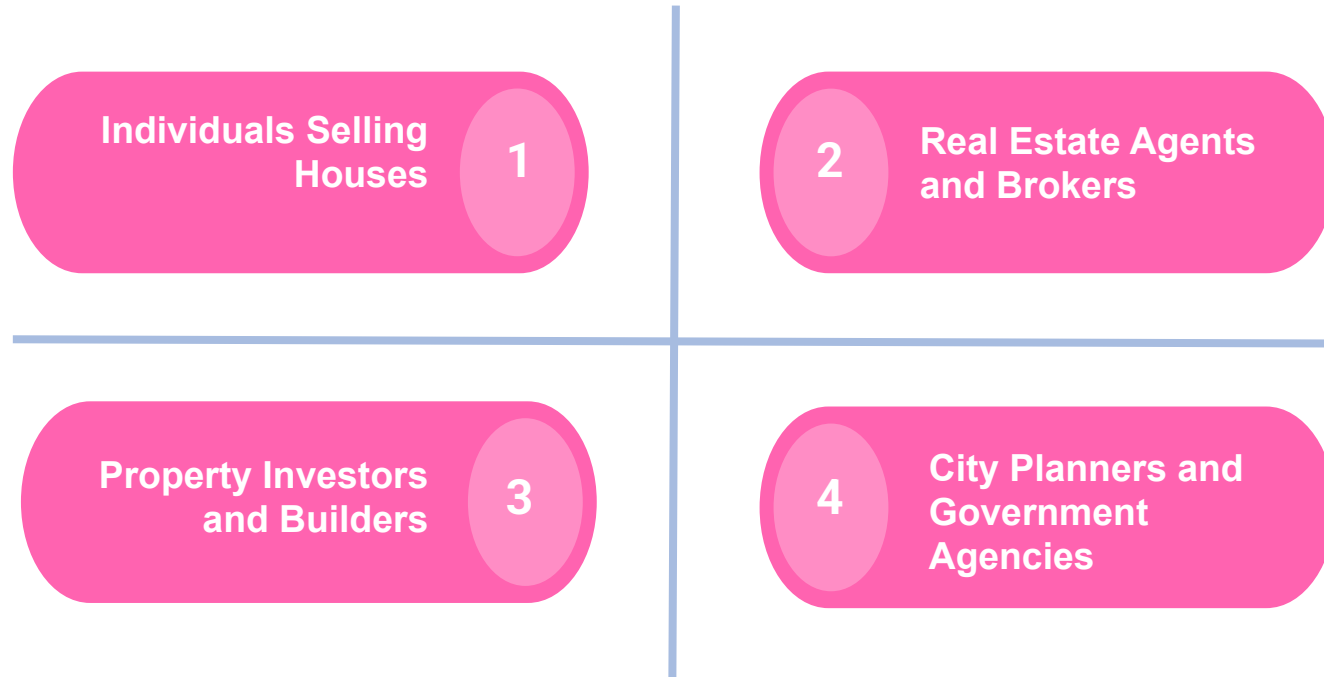
H1: Real-time zip code data on crime rates and housing prices aids informed housing decisions.

H2: Centralized platform is needed for comprehensive Pittsburgh zip code data.

H3: Users embrace digital solution for zip code data despite hesitations.



Buyers



Buyer Persona - Real Estate Agent



SARAH WILSON

REAL ESTATE AGENT

On a lookout for new clients
and selling properties within
Pittsburgh

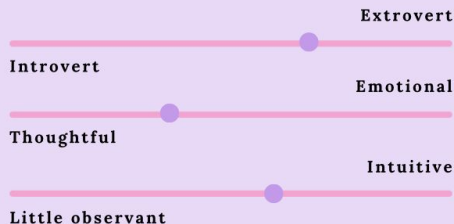
Profile:

- Age: 33
- Gender: **Female**
- Education: **Bachelor's Degree in Business Administration**
- Location: **Pittsburgh, PA**

Goals:

- Finding new clients and the right set of properties best suited for them
- Selling more properties in the neighborhood

Personality:



Challenges:

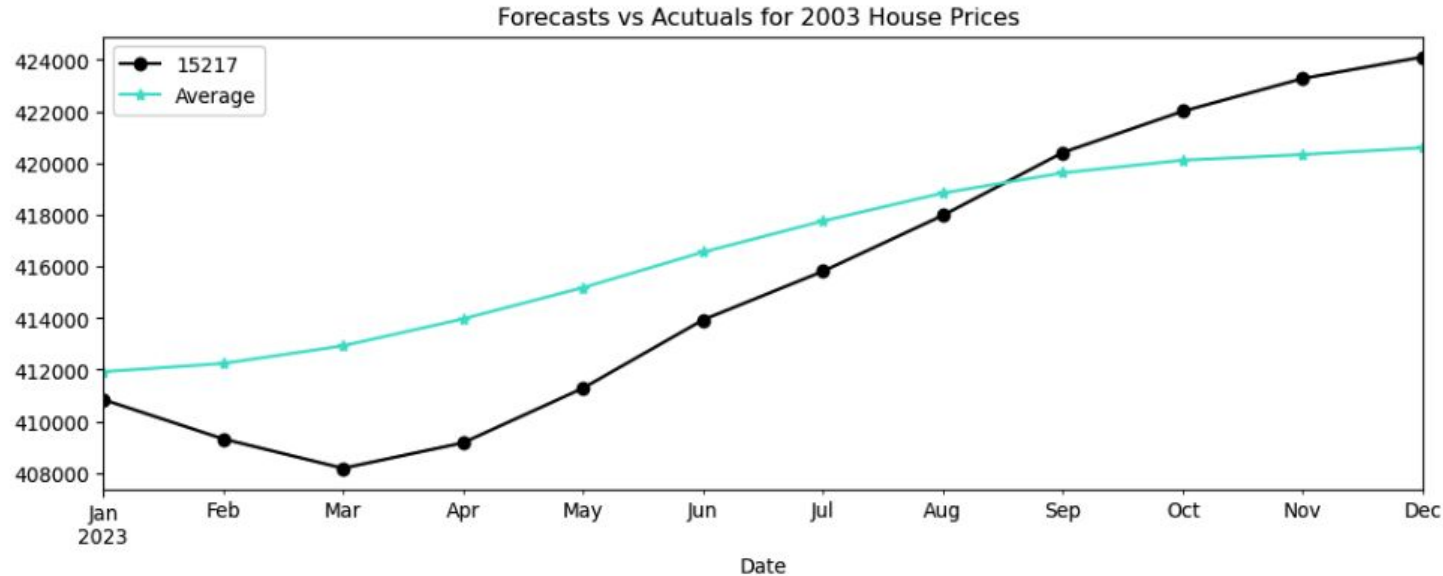
- Keeping up with rapidly changing market conditions
- Managing client expectations and differentiating herself from other real estate agents

Forecasting Techniques

- Applying advanced Times Series Forecasting models
- Extensive Training Period for Machine Learning
- Backtesting Across Four Distinct Test Periods
- Utilizing Cross-Validation to Optimize Model Fit
- Averaging the Performances of the Top Three Models



Forecasts vs Actuals

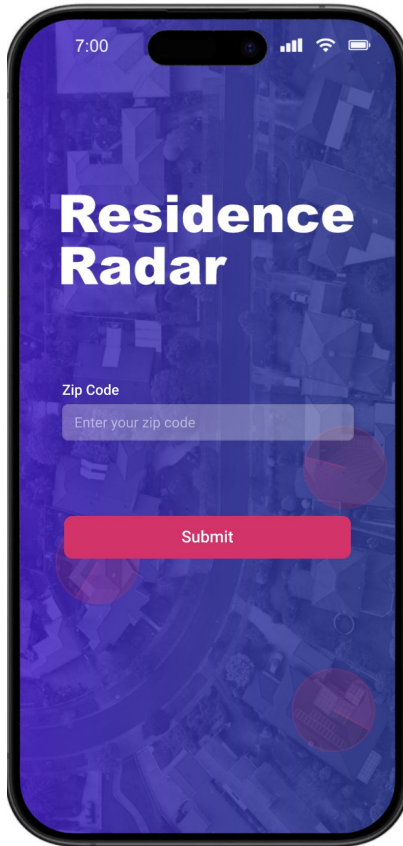


The difference btw the forecasts (in green) and the actual prices (in black) is only around 2 thousands dollars.

Investment Return

In 2023, 4.09 million existing homes were sold in the United States. With 10% of acceptance rate in 3 years, the annual revenue is projected to be about \$800K.

Source	Price	Volume	Revenue
From impressions	\$0.025	400,000	\$10,000
From clicks	\$10	80,000	\$800,000
Total			\$810,000



WireFrame Demo

Appendix - Data Sources

1. “A new app should generally grow by at least 5–10% per week,” [Google source](#)
2. “In 2023, 4.09 million existing homes were sold in the United States, by Statista.com” [Google source](#)
3. \$25 per thousand impressions, [Google source](#)
4. \$10 per click on Zillow, [Google source](#)

